

Gate-State Causal Propagation in a Structured Recurrent Substrate

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Abstract

Structured recurrent systems can expose stable or similar readouts while preserving distinct internal state dynamics. We report Gate-State Causal Propagation, a replicated mechanism discovered in a five-channel recurrent substrate with **fast**, **slow**, **control**, **message**, and **carrier** state. Under a Track B native-emergence protocol, three independent replications passed a mechanism gate: held-out route-disabled and gain-zero divergence remained positive in 3/3 runs, gain-zero diagnostics remained clean, **message** and **carrier** were the top two sensitive channels in 3/3 channel-clamping classifications, and full internal-state resume was exact while surface-only resume left positive gaps. The result suggests that release/gating history embedded in recurrent state can alter downstream dynamics beyond simple additive route injection. A 32-random-control matched comparison shows that the binary gate is not selective by itself, but a stricter Track-B-profile gate separates the positives from most controls: 3/3 Track B positives passed versus 2/32 random unselected controls. This is therefore a replicated Track B phenotype and enriched diagnostic profile, not a sparse-release success or rarity claim. We present the diagnostic procedure, replication artifacts, and limitations, and treat the result as a replicated machine-native discovery lead, enriched under a strict profile gate, motivating Demian v1 as an explicit gate-state design hypothesis.

1 Introduction

Recurrent neural systems are often evaluated through the behavior exposed at their output surface. That is useful, but it is not sufficient for mechanism discovery. Two trajectories can expose similar readouts while carrying different internal state histories, and a stable readout can coexist with structured hidden dynamics. This is especially important when the object of study is not only task performance, but the internal mechanism by which a stateful architecture propagates perturbations, stores history, or changes future dynamics.

Prior work on recurrent networks has shown that dynamical-systems tools can make trained RNN computations interpretable, for example by identifying fixed points, line attractors, and low-dimensional dynamical structure [Maheswaranathan et al., 2019]. Recent work has extended this mechanistic direction to richer RNN behaviors, solution degeneracy across behavior and internal dynamics, and dynamical comparison methods [Torre et al., 2025, Huang et al., 2024, Guilhot et al., 2024]. The present paper uses the same broad motivation but studies a custom structured recurrent substrate: the goal is to determine whether a discovered internal mechanism is causal under ablation and resumable as internal state, not to show benchmark superiority.

The paper is written in a laboratory style. We do not treat a single promoted candidate as a finished theory. Instead, we use discovery runs, ablations, matched controls, and follow-on architectural prototypes as an iterative loop: the substrate produces a candidate internal-state signal, controls test how specific that signal is, and surviving structure becomes a design hypothesis for the next substrate.

We report a compact empirical finding: Gate-State Causal Propagation. In the substrate studied here, candidate systems include a release/gating subsystem that modulates downstream routes. A simple sparse-release interpretation would predict that removing explicit release gain should remove the relevant diagnostic effect. The observed mechanism is different. Across three Track B native-emergence replications, selected candidates retained positive held-out divergence under both route-disabled and gain-zero comparisons, while gain-zero diagnostics remained clean. In the same replications, **message** and **carrier** were repeatedly the most sensitive channels under clamping, and full internal-state resume was exact while surface-only replay failed to resume the trajectory.

The contribution is deliberately narrow:

- a diagnostic framing for gate-state propagation in a structured recurrent substrate;
- a replicated empirical finding across 3/3 Track B native-emergence runs;
- channel-clamping and capsule-continuity evidence supporting the internal-state interpretation;
- a matched-control result showing that the broad binary gate is background-prone but a stricter Track-B-profile gate is enriched over random controls;
- a negative boundary: this is not evidence that sparse delayed release has been solved.

2 Substrate and Terminology

We use *substrate* to mean the recurrent dynamical system being stepped, perturbed, ablated, and compared. A *surface* is the observable readout vector. A *channel* is a named recurrent state component, and a *route* is a transformation between state components. These are operational terms for the experiment, not claims of biological or cognitive status.

The study system is a five-channel extension of the v9 recurrent scaffold. Its state is factorized into:

fast readout-adjacent state;

slow persistent accumulator state;

control lower-dimensional route-control state;

message intermediate routed signal;

carrier slower routed signal supporting delayed propagation.

The scaffold includes a release/gating subsystem that can modulate message/carrier coupling and downstream state updates. The substrate in this paper is the experimental object used to discover and characterize the mechanism. It is not presented as a final architecture.

Capsule continuity is the pause/resume probe used to distinguish surface replay from internal-state continuation. In this paper, a full internal-state capsule means the complete recurrent state needed to resume the trajectory under the current deterministic implementation. It is not a compressed-memory claim.

2.1 Recurrence and ablation notation

For the studied five-channel substrate, write the recurrent state as

$$s_t = (f_t, \ell_t, c_t, m_t, r_t),$$

where f_t is **fast**, ℓ_t is **slow**, c_t is **control**, m_t is **message**, and r_t is **carrier**. The exposed surface is

$$y_t = R_\theta(s_t),$$

and the unablated transition is

$$s_{t+1} = F_\theta(s_t; h_t).$$

Here $h_t = H_\theta(s_t)$ denotes the implicit release/gating variables recomputed from the current recurrent state by the five-channel transition. Because s_t contains the accumulated **message**, **carrier**, **slow**, and **control** trajectory, earlier gating events can affect later h_t only through their effect on the recurrent state. In the v9 five-channel substrate studied here, “gate state” is not a sixth named recurrent channel; it is release/gating history embedded in recurrent state. A later Demian v1 prototype makes this gate state explicit, but that prototype is not the empirical object of this paper.

The main diagnostic transitions are:

$$\begin{aligned} s_{t+1}^{\text{routes}=0} &= F_\theta^{\text{routes}=0}(s_t; h_t), \\ s_{t+1}^{\text{gain}=0} &= F_\theta^{\text{gain}=0}(s_t; h_t), \\ \tilde{s}_{t+1}^{(k)} &= C_k(F_\theta(s_t; h_t)), \end{aligned}$$

where $F_\theta^{\text{routes}=0}$ disables release-mediated route effects, $F_\theta^{\text{gain}=0}$ zeros explicit release gain, and C_k clamps channel k to zero after a transition before feeding the state forward. We use *causal* in this operational ablation sense: a component is treated as causal evidence only when matched interventions change the continuation under this diagnostic protocol.

3 Diagnostic Protocol

3.1 Track B native-emergence search

The candidates reported here come from Track B native-emergence runs, the candidate-promotion protocol used for these native-emergence experiments. Track B is distinct from engineered-target search: instead of selecting only for a predefined sparse or timed release phenotype, it promotes internally useful mechanisms that remain supported under causal diagnostics. This distinction is important because the result reported here is not a sparse-release success. The replicated top candidates are high-duty phenotypes.

3.2 Characterization grid

The focused characterization used seeds 94–102, perturbation scales 0.2, 0.35, and 0.7, two perturbation motifs (**basis:0** and **gaussian:0**), 128 steps, and perturbation step 64. The main focused artifact is the 2026-05-11 gate-state propagation characterization summary in the repository’s `data/diagnostics` directory. The replication artifact is the corresponding 2026-05-11 Track B replication summary.

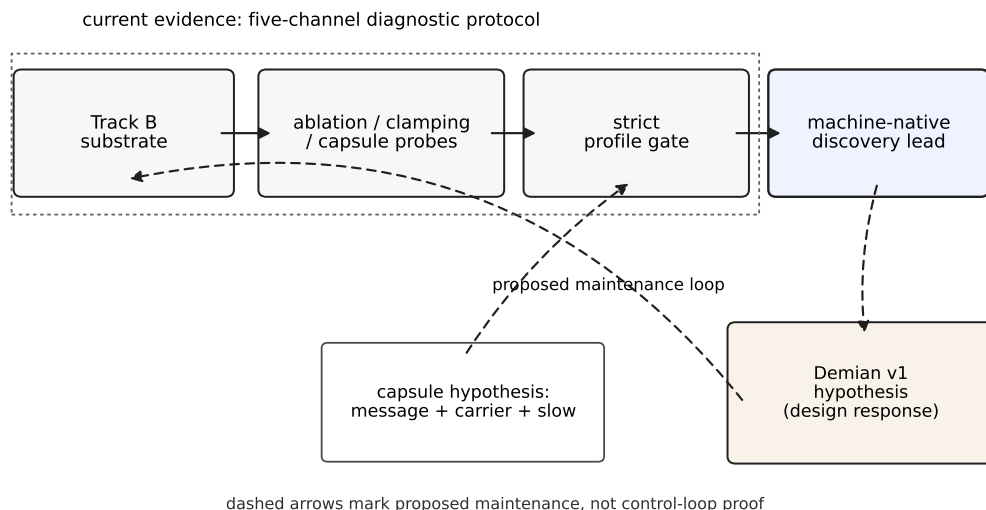


Figure 1: Operational mechanism schematic and proposed maintenance loop. Solid arrows summarize the evidence path tested in the current five-channel diagnostic protocol. The dashed loop is a proposed maintenance loop from the surviving discovery lead into Demian v1 design hypotheses; it is not presented as control-loop proof inside the current diagnostic protocol.

3.3 Interventions

The diagnostic protocol compares the original trajectory against several interventions:

Original the candidate trajectory without diagnostic ablation.

Routes-disabled release-mediated route effects are disabled while the rest of the model remains available.

Gain-zero explicit release gain is set to zero. The diagnostic also checks that gain-zero release-related quantities remain zero within tolerance.

Channel-disabled a named state channel is zeroed after each step and the clamped state is fed into the next step.

The channel-disabled semantics matter. Each step first advances the model; then the requested channel is zeroed; the row is recorded; the clamped state is fed forward. This tests whether continuing with that channel removed changes the trajectory.

3.4 Mechanism gate

Gate-State Causal Propagation is the observed case where prior release/gating history changes later internal evolution, especially through **message**, **carrier**, and **slow** state, even when explicit additive release gain is zero. The mechanism claim is therefore not that a release vector directly injects a signal at the diagnostic step; it is that implicit release/gating state has already shaped the downstream recurrent state so that ablated continuations diverge from the original.

Table 1: Track B Gate-State Causal Propagation replication summary. Values are from `gate.state.track.b.replication.summary.20260511`. Route-disabled and gain-zero are distinct intervention labels in the diagnostic suite; for these held-out summaries they produced identical original-vs-ablated divergence values.

Run	Candidate	Duty	Route div.	Gain-zero div.	Gain-zero clean
1	<code>gen016_candidate012</code>	0.5944	0.2422	0.2422	yes
2	<code>gen019_candidate008</code>	0.6751	0.3194	0.3194	yes
3	<code>gen017_candidate000</code>	0.8906	0.3375	0.3375	yes
Mean	–	–	0.2997	0.2997	3/3

Table 2: Held-out channel-clamping divergences and surface-only resume gaps.

Run	Message	Carrier	Slow	Control	Order	Surface gap
1	1.1289	1.2604	0.8589	0.6411	<code>carrier,message,slow,control</code>	0.6637
2	1.0852	0.8930	0.8101	0.8720	<code>message,carrier,control,slow</code>	0.7789
3	1.0187	1.1618	0.8553	0.6475	<code>carrier,message,slow,control</code>	0.3565

A candidate passes the mechanism gate when route-disabled and gain-zero divergence remain positive, gain-zero cleanliness holds, message/carrier clamping sensitivity repeats, and capsule evidence separates full internal-state resume from surface-only replay. The gate is intentionally stricter than reporting surface behavior alone.

4 Results

4.1 Replication summary

Three Track B native-emergence replications passed the mechanism gate (Table 1). Held-out route-disabled divergence was positive in all three runs. Held-out gain-zero divergence was also positive in all three runs. In this artifact, the mean held-out route-disabled divergence and mean held-out gain-zero divergence are both 0.2997, with per-run values 0.2422, 0.3194, and 0.3375.

The route-disabled and gain-zero divergence columns are identical in the current held-out replication summary because, for these selected candidates and this diagnostic implementation, both interventions expose the same original-vs-ablated divergence value. The equality is reported rather than averaged away because it is a useful validity check: gain-zero cleanliness remains true while the ablated continuation still separates from the original.

The duty-cycle values reinforce the negative boundary of the result. These are not low-duty sparse-release candidates; they are high-duty native-emergence phenotypes that nonetheless preserve a replicated gate-state mechanism under the current diagnostics.

4.2 Channel-clamping sensitivity

The channel-disabled diagnostics identify `message` and `carrier` as the repeated top channels under the clamping intervention (Table 2). They were the top two channels in 3/3 replications. The order varied: `carrier,message,slow,control` in runs 1 and 3, and `message,carrier,control,slow` in run 2. `slow` support was also high across the held-out classifications.

Table 3: Reduced matched control/null comparison from `gate_state_control_nulls_20260515`. The binary gate is broad; the strict descriptive profile adds Track-B-calibrated effect-size and channel-dominance criteria.

Family	N	Binary pass	Strict pass	Route div.	M/C dom.
Track B positives	3	3/3	3/3	0.2994	0.3104
Default unselected	1	1/1	0/1	0.1528	0.3597
Random unselected	32	15/32	2/32	0.1398	0.2033

The repeated message/carrier pattern is the strongest channel-level evidence for the mechanism under this intervention. Because clamping a channel to zero is a strong out-of-distribution intervention, we read these values as channel-clamping sensitivity under the reported protocol, not as a general proof that no other representation could carry the same function. The pattern still suggests that the effect is not localized only in the readout-adjacent **fast** surface, nor explained solely by the persistent **slow** accumulator.

4.3 Gain-zero and route-disabled behavior

The gain-zero diagnostic is designed to distinguish a simple additive release effect from a more stateful gate-history effect. In the focused characterization, gain-zero cleanliness is true: release strength, route norm, and endogenous release norm are zero under the gain-zero condition within the recorded tolerance. Yet original-vs-gain-zero divergence remains positive and matches the route-disabled divergence in the held-out replication rows.

This pattern motivates the name Gate-State Causal Propagation. The observed effect is not that a nonzero release vector directly injects a signal under gain-zero. Rather, the candidate’s release/gating history embedded in recurrent state appears to alter later dynamics in a way that remains visible under the diagnostic comparison. The result should therefore be separated from a route-specific sparse-release claim.

4.4 Reduced matched control comparison

We added a matched control/null comparison using the same route-disabled, gain-zero, channel-clamping, and capsule criteria. The reduced grid used seeds 96–98, perturbation scale 0.35, motifs `basis:0` and `gaussian:0`, and the same seven intervention conditions. The binary gate was not selective by itself: the default unselected control passed in 1/1 case and random unselected controls passed in 15/32 cases. We therefore report a stricter Track-B-calibrated descriptive profile, using the weakest Track B positive as the threshold. This profile is not a pre-registered discovery criterion; it is a post hoc effect-size summary used to test whether the replicated positives occupy a more selective region than random unselected controls. It requires the binary gate, route/gain-zero divergence at least as large as the weakest Track B positive, and message/carrier clamping dominance over slow/control at least as large as the weakest Track B positive. Under that stricter profile, Track B positives passed in 3/3 cases, the default unselected control passed in 0/1, and random unselected controls passed in 2/32 (Table 3). The Track B versus random enrichment factor was 16.0, with one-sided Fisher exact $p = 0.00153$.

This control result changes the interpretation. The Track B candidates remain replicated positives under the diagnostic protocol, but the original binary gate cannot support a rarity or discovery-specificity claim. The stricter descriptive profile gives a stronger artifact: it sharply reduces random-control passes while preserving all three Track B positives. Because the profile is

post hoc, its remaining false positives show that larger null populations, pre-specified thresholds, and independent replications are still required before treating the profile as rare.

4.5 Capsule continuity

Surface-only replay was insufficient to resume the trajectory. Full internal-state resume was exact in 3/3 replications, as expected for a deterministic full-state continuation, while surface-only resume left a positive final gap in 3/3. The surface-only gaps were 0.6637, 0.7789, and 0.3565. The focused characterization also reports exact full-internal-state resume over 54 capsule runs for the representative candidate: full internal state and uninterrupted continuation both have zero final L2 gap and zero mean-step gap, while surface-only replay has positive final and mean-step gaps.

This result should be read narrowly. It shows that, under this deterministic probe, the complete internal state resumes the trajectory while surface-only replay does not. It does not show that the state has been compressed, nor that the capsule is minimal.

5 Discussion

The result supports an internal-state-centered view of this substrate. The release/gating subsystem is not only an output-side switch; its history can be part of the downstream computation. The repeated sensitivity of `message` and `carrier` under channel clamping further suggests that the mechanism depends on routed intermediate state rather than on the readout surface alone.

The capsule-continuity result gives a complementary test of surface insufficiency. Full-state resume is expected to be exact in a deterministic RNN-style system, so the informative comparison is the failure of surface-only replay. If surface state were sufficient for this computation under the probe, surface-only replay should resume the trajectory as well as full internal-state continuation. It does not. The stronger conclusion is not that full-state copying is a useful compression method, but that evaluation based only on the exposed readout would miss the state needed for continuation.

This result also motivates an architectural response: make gate state explicit in Demian v1. The current Demian v1 prototype exists as a six-channel architecture with `fast`, `slow`, `control`, `message`, `carrier`, and `gate` state. Its gate channel modulates message-carrier, carrier-slow, and message/carrier-surface routes without injecting an additive release vector. That prototype is a design response with unit-level architectural tests, not a validated empirical result of this paper. The present evidence supports a narrower claim: in the current v9 five-channel Track B protocol, gate-state propagation replicated and survived the reported ablation and resume tests.

This is the sense in which we treat the result as a discovery lead rather than a closed discovery claim. The lead was not supplied by a human benchmark target or by an externally imposed human-facing objective. It emerged from a machine-native state trajectory, survived a first replication and ablation round, weakened under broad controls, and then sharpened into an enriched message/carrier-dominant profile. Demian v1 is therefore framed as a design response to the substrate’s own diagnostic signal, not as proof that the signal has already been fully explained.

As a lab interpretation, Demian is best understood here as an experimental substrate workflow, not just a model instance. The workflow discovers native signals in a structured recurrent substrate, applies ablations and controls to separate robust internal-state structure from surface coincidence, and then turns the surviving profile into the next design hypothesis. In the lab timeline, related probes moved from transformer-style state inspection through Mamba-like sequence-state experiments and then into explicit recurrent substrates such as vanilla RNN, GRU, and LSTM-style scaffolds; the present five-channel substrate is the point where the message/carrier/slow capsule

hypothesis became experimentally inspectable. In this paper, that means a replicated machine-native discovery lead, enriched under the strict Track-B-profile gate, motivates Demian v1 without validating it.

6 Limitations

The result is limited to the current substrate family and current Track B protocol. It has not yet been shown to be universal across recurrent architectures, independent implementations, or broader search spaces. The replicated candidates are high-duty phenotypes, so this paper does not validate the original sparse delayed-release target.

The current evidence is also selected-candidate evidence. The reduced matched control comparison shows that the binary mechanism gate has high background pass rates in unselected controls, while the stricter profile gate reduces but does not eliminate random-control passes. The capsule evidence is full-state evidence: it shows surface insufficiency under this probe, not a minimal or compressed memory. Broader tests should use larger matched control populations, stronger held-out sets, and independent reimplementations of the diagnostic protocol.

7 Reproducibility

The public repository is:

```
https://github.com/Aeshma-Daeva/Demian
```

The compact local test path for the core Gate-State result is:

```
./venv/bin/python -m pytest \
  tests/test_gate_state_propagation_characterization.py \
  tests/test_v9_capsule_continuity.py \
  tests/test_gate_state_control_nulls.py -q
```

The main replication artifact can be inspected with:

```
./venv/bin/python - <<'PY'
import json
from pathlib import Path

summary = json.loads(Path(
    "data/diagnostics/gate_state_track_b_replication_summary_20260511/"
    "summary.json"
).read_text())
print(json.dumps({
    "replication_count": summary.get("replication_count"),
    "passed_replication_count": summary.get("passed_replication_count"),
    "message_carrier_top2_count": summary.get("message_carrier_top2_count"),
    "mean_heldout_routes_divergence": summary.get(
        "mean_heldout_routes_divergence"
    ),
    "mean_heldout_gain_zero_divergence": summary.get(
```



```

        "mean_heldout_gain_zero_divergence"
    ),
}, indent=2, sort_keys=True))
PY

```

Primary artifacts under `data/diagnostics`:

- `gate_state_propagation_characterization_20260511/summary.json`
- `gate_state_track_b_replication_summary_20260511/summary.json`
- `gate_state_track_b_replication_1_heldout_20260511/summary.json`
- `gate_state_track_b_replication_2_heldout_20260511/summary.json`
- `gate_state_track_b_replication_3_heldout_20260511/summary.json`
- `gate_state_control_nulls_20260515/control_null_summary.json`
- `../substrate_lab/v9_capsule_continuity_20260511/summary.json`

Exact paths are listed in the repository artifact index and manifest.

8 Conclusion

Gate-State Causal Propagation is a replicated internal mechanism in the current structured recurrent substrate protocol. Across three Track B replications, the mechanism preserved positive held-out route-disabled and gain-zero divergence, clean gain-zero diagnostics, repeated message/carrier channel-clamping sensitivity, and exact full internal-state resume with positive surface-only resume gaps.

The result is intentionally bounded. It is not a sparse-release success claim, not a benchmark-superiority claim, not a final rarity claim, and not a validation of a final architecture. Its value is narrower and more concrete: it identifies a replicated internal-state diagnostic pattern, shows that a strict Track-B-calibrated descriptive profile is enriched over random unselected controls, and turns a replicated machine-native discovery lead into a testable architectural motivation for Demian v1 and later gate-state experiments.

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